



CalCOFI

The purpose of this research paper is to provide a comprehensive, step-by-step guide for conducting a **Repeated Measures ANOVA** using the Isalos Analytics Platform. Repeated Measures ANOVA extends one-way ANOVA by analyzing **multiple measurements** taken from the same subjects (or casts) across different conditions or time points, allowing analysts to assess how within-subject factors (e.g., depth) and between-subject factors (e.g., cast) influence a set of interrelated continuous outcomes. This methodology is particularly valuable in oceanographic research, where environmental parameters such as temperature and salinity are measured at various depths and should be analyzed together to account for within-cast correlations.

Repeated Measures ANOVA combines the principles of multivariate ANOVA (MANOVA) and univariate repeated measures. By including within-subject factors (e.g., Depth), the procedure accounts for the correlation between repeated measurements, thereby increasing statistical power and providing more accurate estimates of differences across conditions while controlling for individual variability.

Isalos version used: 2.1.5

The dataset used in this analysis is the **CalCOFI (California Cooperative Oceanic Fisheries Investigations) reduced Dataset**, available on Kaggle.

Source: [CalCOFI](#)

This dataset simulates real-world oceanographic records and includes cast-level information such as bottle count, cast count, depth measurements, temperature, and salinity. It is designed to support environmental monitoring and ecological research.

For the purposes of this Repeated Measures ANOVA analysis, variables are classified as follows:

- **Dependent Variables (Responses):**

- *T_degC* — Temperature in degrees Celsius or
- *Salnty* — Salinity in practical salinity units

These are continuous outcome measures to be analyzed simultaneously across depths.

- **Within-Subject Factor:**

- *Depthm* — Depth in meters (e.g., 0, 10, 20, 30, 50, 75 m)

This categorical variable represents the repeated-measures condition across which each cast is measured.

- **Between-Subject Factor:**

- *Cst_Cnt* — Cast count (e.g., 1 through 6)

This categorical variable represents independent groups whose multivariate effect on the oceanographic outcomes is of primary interest.

Step 1: Import Data from file

Firstly, launch the Isalos Analytics Platform (version 2.1.2). Then, in the left spreadsheet panel, create a new tab by clicking **File** and choose **Open** to load the CalCOFI dataset (or simply copy-paste the data from your spreadsheet).

Btl_Cnt	Cst_Cnt	Depthm	T_degC	Salnty
1	1	0	10.5	33.44
2	1	10	10.46	33.437
3	1	20	10.45	33.421
4	1	30	10.45	33.431
5	1	50	10.24	33.424
6	1	75	9.86	33.494
7	1	100	9.67	33.58
8	2	0	10.1	32.95
9	2	10	9.89	32.94
10	2	20	9.75	32.939
11	2	30	9.52	33.013
12	2	50	9.04	33.272
13	2	75	8.83	33.488
14	2	100	8.26	33.631
15	3	0	10.2	32.63
16	3	10	10.05	32.66

Step 2: Access the RM-ANOVA function

Navigate to the analysis module through the main menu by choosing the following path:

Statistics → Analysis of (Co)Variance → RM ANOVA

Then, you will see the RM ANOVA configuration window:

Repeated Measures ANOVA

Confidence Level (%) Double (0,100), Default: -

Sum of Squares for Tests Adjusted (Type III)

Subject Column

Within Column

Dependent Variable

Execute Cancel

Step 3: Configure RM ANOVA parameters

The Repeated Measures ANOVA model can be specified with varying levels of complexity. The model is expressed as:

$$[T_degC, Salnty] = \mu + \alpha_i(Cst_Cnt) + \beta_j(Depthm) + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

This is a multivariate repeated measures model where the two dependent variables are regressed onto the within-subject factor (Depth) and the between-subject factor (Cast). The model includes main effects and the interaction between Cast and Depth.

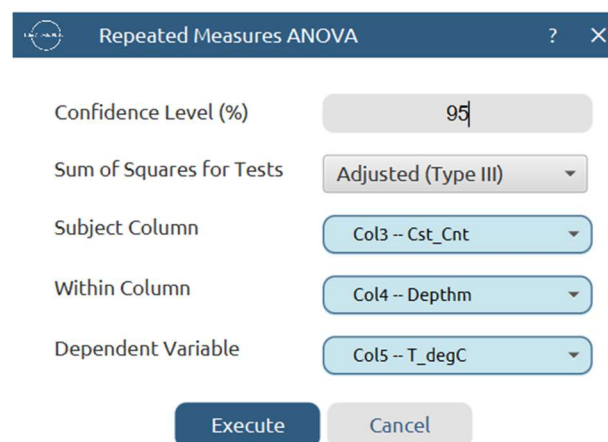
Important: The inclusion of interaction terms (e.g., $Cst_Cnt \times Depthm$) should be guided by theoretical considerations and exploratory data analysis. Repeated Measures ANOVA assumes sphericity—if this assumption is violated, corrections (e.g., Greenhouse-Geisser) may be necessary. For this basic demonstration, we focus on the **full factorial model**.

Steps in Isalos Analytics Platform:

In the RM ANOVA configuration window, set the following parameters:

- * **Confidence Level (%)**: 95
- * **Subject Column**: Col3 -- Cst_Cnt
- * **Within Column**: Col4 -- Depthm
- * **Dependent Variable**: Col5 -- T_degC

The formula preview will display the multivariate model as shown below:



The screenshot shows a dialog box titled "Repeated Measures ANOVA" with a close button (X) and a help button (?). The dialog contains five configuration fields:

- Confidence Level (%): A text input field containing "95".
- Sum of Squares for Tests: A dropdown menu showing "Adjusted (Type III)".
- Subject Column: A dropdown menu showing "Col3 -- Cst_Cnt".
- Within Column: A dropdown menu showing "Col4 -- Depthm".
- Dependent Variable: A dropdown menu showing "Col5 -- T_degC".

At the bottom of the dialog are two buttons: "Execute" (highlighted in blue) and "Cancel".

After configuring all parameters:

- * Review all selections to ensure accuracy.
- * Click the **Execute** button to run the analysis.
- * Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

WITHIN-SUBJECTS EFFECTS							
Source	Depthm	DF	Adj SS	Adj MS	F-Value	P-Value	Significant
Depthm	Linear	1	6.5331175	6.5331175	21.3491685	0.0012534	Yes
Error(Depthm)	Linear	9	2.7541146	0.3060127			
Depthm	Quadratic	1	1.1805001	1.1805001	12.8985658	0.0058255	Yes
Error(Depthm)	Quadratic	9	0.8236963	0.0915218			
Depthm	Cubic	1	0.0601667	0.0601667	1.1751302	0.3065340	No
Error(Depthm)	Cubic	9	0.4608000	0.0512000			
Depthm	Quartic	1	0.0056894	0.0056894	0.2649781	0.6191230	No
Error(Depthm)	Quartic	9	0.1932392	0.0214710			
Depthm	Order 5	1	0.0001458	0.0001458	0.0345801	0.8566020	No
Error(Depthm)	Order 5	9	0.0379554	0.0042173			
Depthm	Order 6	1	0.0043091	0.0043091	2.7022689	0.1346180	No
Error(Depthm)	Order 6	9	0.0143516	0.0015946			
BETWEEN-SUBJECTS EFFECTS							
Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant	
Intercept	1	8260.1165714	8260.1165714	892.4651812	0E-7	Yes	
Error	9	83.2985429	9.2553937				

Step 5 : Residual Analysis & Assumption's checks

Choose

Statistics → Analysis of (Co)Variance → RM ANOVA

and choose

Residual Analysis & Assumption's Checks.

You get the following window:

Residual Analysis & Assumption's Checks

Normality of Residuals

Statistical Tests

- ☐ Shapiro-Wilk Test
- ☐ Anderson-Darling Test
- ☐ Kolmogorov-Smirnov Test

Plots

- ☐ Normal Q-Q Plots
- ☐ Histogram of Residuals
- ☐ Density Plot of Residuals

Homogeneity of Variance
(Homoscedasticity)

Statistical Tests

- ☐ Mauchly's Test

Plots

- ☐ Residuals vs Fitted Values
- ☐ Residuals vs Dependent Variable
- ☐ Boxplots by Group
- ☐ Violin Plot by Group

Residual Diagnostics

Plots

- ☐ Residuals vs Observation Order

Model Adequacy

Plots

- ☐ Residuals vs Fitted values
- ☐ Residuals vs Each Factors

Save

Cancel

and then choose the tests you want. For example, choose the following:

Residual Analysis & Assumption's Checks

?

Normality of Residuals

Statistical Tests

☒

 Shapiro-Wilk Test

☒

 Anderson-Darling Test

☒

 Kolmogorov-Smirnov Test

Plots

☒

 Normal Q-Q Plots

☒

 Histogram of Residuals

☒

 Density Plot of Residuals

Residual Diagnostics

Plots

☒

 Residuals vs Observation Order

Homogeneity of Variance
(Homoscedasticity)

Statistical Tests

☐

 Mauchly's Test

Plots

☒

 Residuals vs Fitted Values

☒

 Residuals vs Dependent Variable

☒

 Boxplots by Group

☒

 Violin Plot by Group

Model Adequacy

Plots

☒

 Residuals vs Fitted values

☒

 Residuals vs Each Factors

Save

Cancel

and click save. You should have the following window open:


Repeated Measures ANOVA
?

Confidence Level (%)

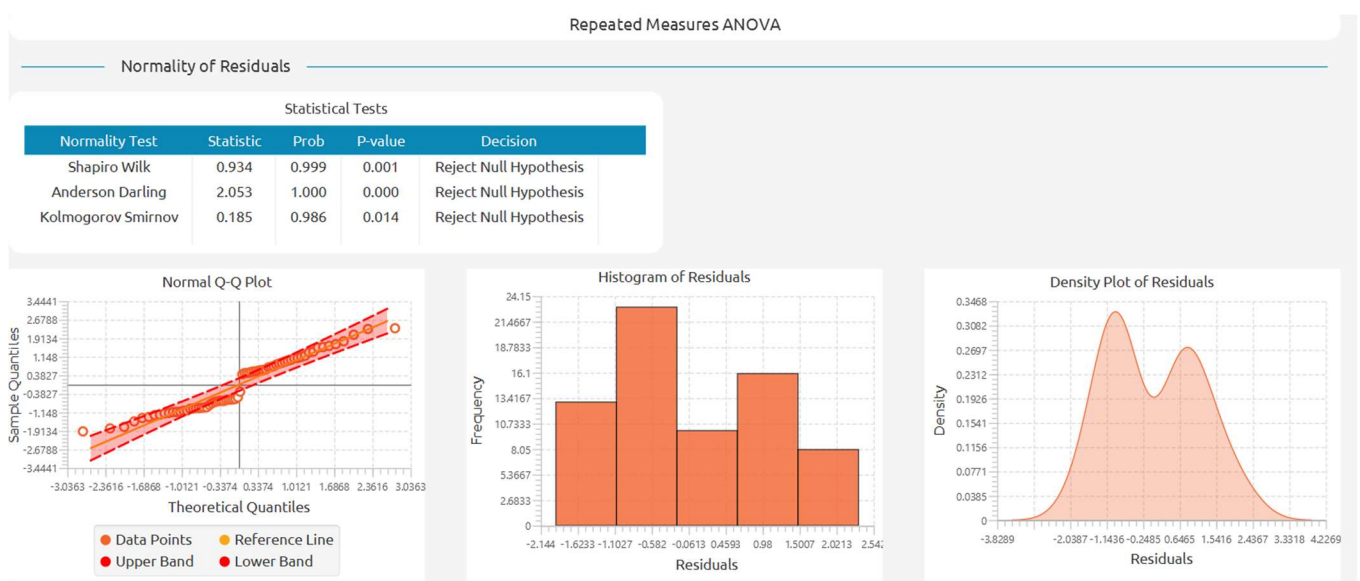
Sum of Squares for Tests

Subject Column

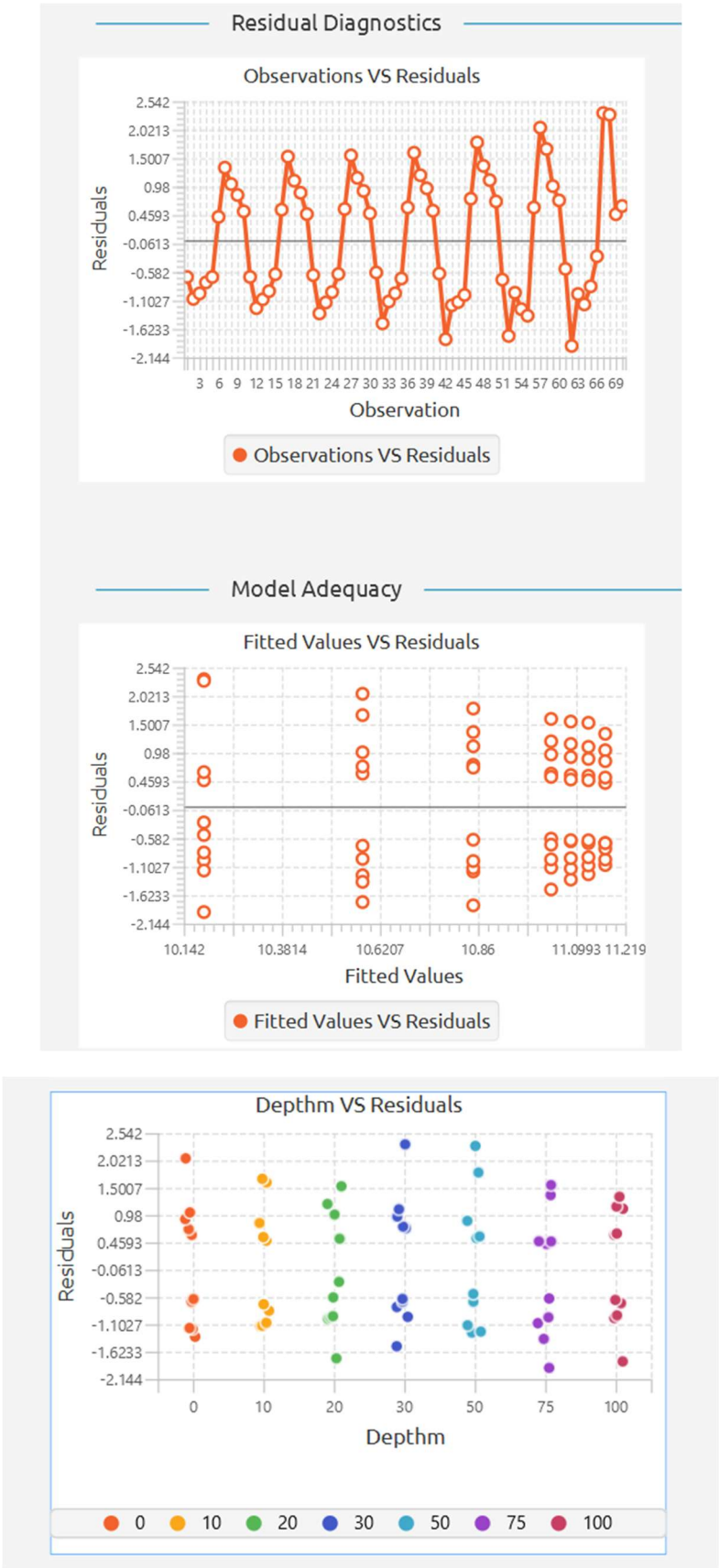
Within Column

Dependent Variable

After that, click execute and you get the results







Part 2: Descriptive Statistics

This step clarifies the central tendency, variability, and distribution of temperature and salinity across depths and casts. It also helps to detect any unusual values, check the balance of the design, and provide a clear context for the inferential tests that follow.

In this section, we present means, standard deviations, and minimum/maximum values for the two dependent variables – **T_degC** (temperature) and **Salnty** (salinity) – grouped by **Depthm** (within-subject factor) and by **Cst_Cnt** (between-subject factor). We also provide a cross-tabulation of means for each combination of depth and cast to reveal interaction patterns.

Step 1: Import the data (use the same data as before)

Step 2: Access the descriptive statistics function

Navigate to the descriptive analysis module through the main menu by choosing the following path:

Statistics → Basic Statistics → Descriptive Statistics

Then, you will see the descriptive statistics configuration window:

 Descriptive Statistics
 ? ×

Select Type Sample

1. Sample Size & Counts

☐ Size/Count (n)
 ☐ Frequency
 ☐ Sum (Σ)
 ☐ Sum of Squares (SS)

2. Central Tendency

☐ Mean ($\bar{x}$)
 ☐ Median ($\tilde{x}$)
 ☐ Mode
 ☐ Midrange (MR)

3. Dispersion / Spread

☐ Standard Deviation (s)
 ☐ Variance (s^2)
 ☐ Range (R)
 ☐ Mean Absolute Deviation (MAD)
 ☐ Minimum (min)
 ☐ Maximum (max)

4. Quartiles, UQR & Outliers

Method: ☐ Inclusive ☐ Exclusive
 ☐ Quartiles (Q1, Q2, Q3)
 ☐ Interquartile Range (IQR)
 ☐ Outliers

5. Relative / Advanced Measures

☐ Coefficient of Variation (CV)
 ☐ Relative Standard Deviation (RSD)
 ☐ Root Mean Square (RMS)
 ☐ Standard Error of the Mean (SE $\bar{x}$)

6. Distribution Shape

☐ Skewness (γ_1)
 ☐ Kurtosis (β_2)
 ☐ Kurtosis Excess (α_2)

7. Normality Tests

☐ Shapiro-Wilk
 ☐ Kolmogorov-Smirnov
 ☐ Anderson-Darling

8. Plots

☐ Q-Q Plot
 ☐ Histogram
 ☐ Box Plot
 ☐ Violin Plot

Execute

Cancel

Step 3: Configure parameters

In the descriptive statistics window, set the following parameters:

1. Sample Size & Counts
 - a. Size / Count (n)
 - b. Frequency
 - c. Sum
 - d. Sum of Squares
2. Central Tendency
 - a. Mean
 - b. Median
 - c. Mode
 - d. Midrange

3. Dispersion / Spread
 - a. Standard Deviation
 - b. Variance
 - c. Range
 - d. Mean Absolute Deviation
 - e. Minimum
 - f. Maximum

4. Quartiles, UQE & Outliers
 - a. Quartiles (Q1,Q2,Q3)
 - b. Interquartile Range (IQR)
 - c. Outliers

5. Relative / Advanced Measures
 - a. Coefficient of Variation
 - b. Relative Standard Deviation
 - c. Root Mean Square
 - d. Standard Error of the Mean

6. Distribution Shape
 - a. Skewness
 - b. Kurtosis
 - c. Kurtosis excess

7. Normality Tests
 - a. Shapiro-Wilk
 - b. Kolmogorov-Smirnov
 - c. Anderson-Darling

8. Plots
 - a. Q-Q Plot
 - b. Histogram
 - c. Box Plot
 - d. Violin Plot

The above preview will display the multivariate model as shown below:

Descriptive Statistics
?
×

Select Type
Sample

1. Sample Size & Counts
☒ Size/Count (n)
☒ Frequency
☒ Sum (Σ)
☒ Sum of Squares (SS)

2. Central Tendency
☒ Mean ($\bar{x}$)
☒ Median ($\tilde{x}$)
☒ Mode
☒ Midrange (MR)

3. Dispersion / Spread
☒ Standard Deviation (s)
☒ Variance (s^2)
☒ Range (R)
☒ Mean Absolute Deviation (MAD)
☒ Minimum (min)
☒ Maximum (max)

4. Quartiles, UQR & Outliers
Method: ☒ Inclusive ☒ Exclusive
☒ Quartiles (Q1, Q2, Q3)
☒ Interquartile Range (IQR)
☒ Outliers

5. Relative / Advanced Measures
☒ Coefficient of Variation (CV)
☒ Relative Standard Deviation (RSD)
☒ Root Mean Square (RMS)
☒ Standard Error of the Mean (SE $\bar{x}$)

6. Distribution Shape
☒ Skewness (γ_1)
☒ Kurtosis (β_2)
☒ Kurtosis Excess (α_2)

7. Normality Tests
☒ Shapiro-Wilk
☒ Kolmogorov-Smirnov
☒ Anderson-Darling

8. Plots
☒ Q-Q Plot
☒ Histogram
☒ Box Plot
☒ Violin Plot

Distribution: Normal

Execute
Cancel

After configuring all parameters:

- * Review all selections to ensure accuracy.
- * Click the **Execute** button to run the descriptive analysis.
- * Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

Factors	Size/Count	Sum	Sum of Squares	Mean	Median	Mode	Mid Range	Standard Deviation	Variance	Range
Btl_Cnt	70.0	2485.0	28577.5	35.5	35.5	None	35.5	20.3510851	414.1666667	69.0

Mean Absolute Deviation	Minimum	Maximum	Quartiles - Inclusive	Quartiles - Exclusive	Interquartile Range - Inclusive	Interquartile Range - Exclusive
17.5	1.0	70.0	Q1: 18.250000, Q2: 35.500000, Q3: 53.750000	Q1: 17.750000, Q2: 35.500000, Q3: 53.750000	34.5	35.5

Anderson-Darling: A ²	Anderson-Darling: p-value	Anderson-Darling: Prob	Anderson-Darling: Decision	Kolmogorov-Smirnov: D	Kolmogorov-Smirnov: p-value	Kolmogorov-Smirnov: Prob	Kolmogorov-Smirnov: Decision
0.0626790	0.9299943	0.0700057	Fail to Reject Null Hypothesis	0.7530347	0.0477094	0.9522906	Reject Null Hypothesis

Coefficient Of Variation	Relative Standard Deviation	Root Mean Square	Standard Error Of Mean	Skewness	Kurtosis	Kurtosis Excess	Shapiro-Wilk: W	Shapiro-Wilk: p-value	Shapiro-Wilk: Prob	Shapiro-Wilk: Decision
0.5732700	57.3270004	40.847277	2.4324199	0E-7	1.8000000	-1.2000000	0.9550122	0.0133838	0.9866162	Reject Null Hypothesis

